



# Ensemble Methods

*Under what conditions does boosting outperform bagging, and vice versa — and why?*

## *Boosting vs. Bagging*

From-scratch implementation of Decision Trees, AdaBoost, and Random Forest, with an empirical comparison supported by PCA / K-Means / DBSCAN analysis

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AI Academy · National AI Center  
July 2026

# Team Contributions



**Ferman Khankishiyev**

Team Lead

CI  
Decision Tree  
Testing, code quality  
Multiclass part for AdaBoost  
Metrics(f1, auc-roc, accuracy)  
Head-to-Head Comparison experiment  
Baseline experiment



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Developer Lead

AdaBoost  
Preprocessing  
DT&RF-optimization  
Exploratory Data Analysis  
Adaboost scaling experiment  
Noise Robustness experiment



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Presentation  
Random Forest  
Random Forest scaling experiment  
Bias-variance decomposition experiment



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Documentation Lead  
Developer Support

Presentation  
Report  
PCA/DBSCAN/K-Means  
Bias-variance Decomposition  
Unsupervised Analysis



# Three Models Implemented From Scratch



## 1 · Decision Tree

CART methodology, Gini / Entropy criterion, depth-based pruning, exhaustive search over continuous features.



## 2 · AdaBoost (SAMME)

Depth-1 decision stumps as weak learners, sample-weight update rule, staged predictions.



## 3 · Random Forest

Bootstrap aggregating, random feature subsampling, out-of-bag (OOB) accuracy, parallel training.

*sklearn's DecisionTreeClassifier / AdaBoostClassifier / RandomForestClassifier are used only as a sanity-check baseline.*



# Decision Tree — CART Algorithm

## Splitting Criterion

### Gini impurity

$$I_G(p) = 1 - \sum_{c=1}^C p_c^2$$

### Entropy

$$I_E(p) = - \sum_{c=1}^C p_c \cdot \log_2(p_c)$$

### Impurity reduction

$$\Delta I = I_{\text{parent}} - N_{\text{left}} / N \cdot I_{\text{left}} - N_{\text{right}} / N \cdot I_{\text{right}}$$

Stopping conditions: max\_depth reached, min\_samples\_split violated, a pure node, or identical feature vectors.

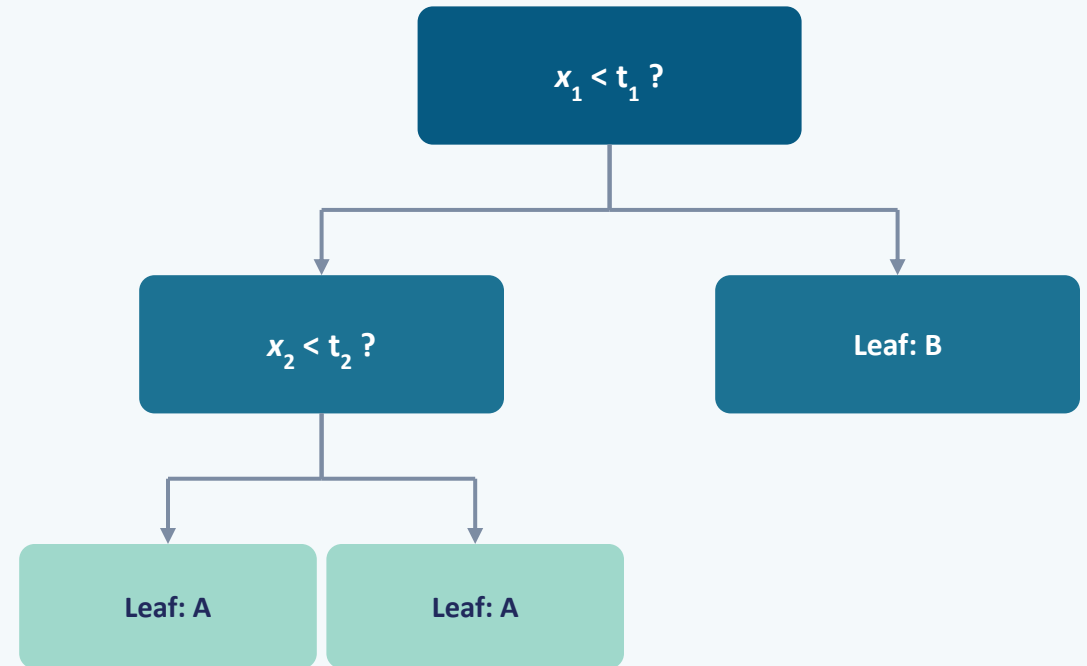


Figure: a two-level decision-tree example



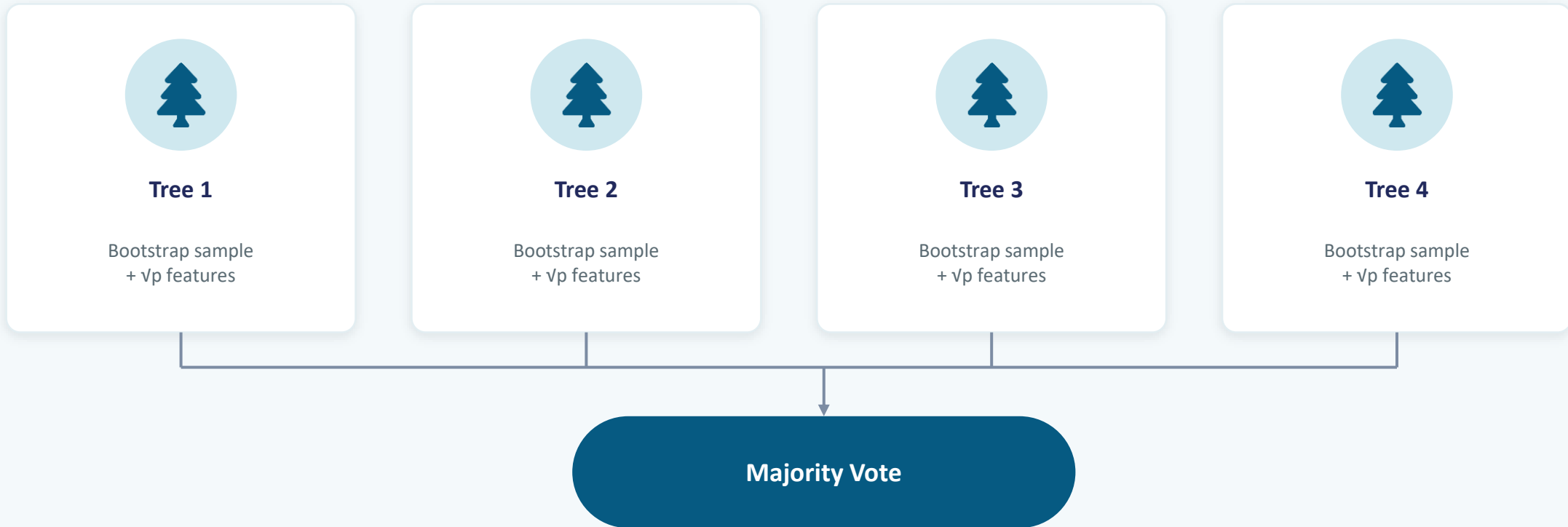
# AdaBoost — Weight Update Rule

- 1 Initialize equal weights:  $w_i = 1 / N$
- 2 Fit a weak learner (stump)  $h_m$  on the weighted data
- 3 Compute weighted error:  $\varepsilon_m = \sum w_i \cdot 1[h_m(x_i) \neq y_i] / \sum w_i$
- 4 Model weight:  $\alpha_m = \ln((1 - \varepsilon_m) / \varepsilon_m)$
- 5 Update sample weights:  $w_i \leftarrow w_i \cdot \exp(\alpha_m \cdot 1[\text{error}])$ , then normalize
- 6 Final prediction:  $\hat{y} = \text{argmax}_k \sum \alpha_m \cdot 1[h_m(x) = k]$

If  $\varepsilon_m \geq 0.5$  the algorithm stops; if  $\varepsilon_m = 0$  it is clipped to  $10^{-10}$ .



# Random Forest — Bagging + Feature Subsampling



**OOB (Out-of-Bag) evaluation:** for each sample, predictions are pooled only from the trees that never saw it — model generalization is measured without a separate test set.

*Feature importances: averaged impurity reduction across all trees.*



# Unsupervised Analysis Pipeline



## PCA

Scree plot, cumulative explained variance, projection onto the first 2 components.



## K-Means

Elbow method ( $k = 1-10$ ), 10 restarts, Adjusted Rand Index (ARI).



## DBSCAN

k-distance plot, core/noise point classification, ARI and noise fraction.

*2D PCA scatter plots colored by true class, K-Means clusters, and DBSCAN clusters are compared side by side.*

# Baseline



- Train a single unpruned DecisionTree on an 80/20 train/test split.
- Train a depth-1 stump DecisionStump.
- Train sklearn.tree.DecisionTreeClassifier with identical parameters.
- Report accuracy, F1-macro, and AUC-ROC for binary or macro-averaged AUC for multi-class.

## 1) Breast Cancer

| Model               | Accuracy | F1-score | AUC    |
|---------------------|----------|----------|--------|
| DecisionTree (ours) | 0.9561   | 0.9539   | 0.9468 |
| DecisionStump       | 0.8684   | 0.8579   | 0.8468 |
| sklearn Tree        | 0.9386   | 0.9364   | 0.9351 |

Accuracy difference vs sklearn: 1.87% (PASS)

## 2) Adult Income

| Model               | Accuracy | F1-score | AUC    |
|---------------------|----------|----------|--------|
| DecisionTree (ours) | 0.7783   | 0.6981   | 0.6997 |
| DecisionStump       | 0.8033   | 0.6027   | 0.5945 |
| sklearn Tree        | 0.7763   | 0.6969   | 0.6996 |

Accuracy difference vs sklearn: 0.25% (PASS)

## 3) Covertypes

| Model               | Accuracy | F1-score | AUC    |
|---------------------|----------|----------|--------|
| DecisionTree (ours) | 0.9178   | 0.8733   | 0.9265 |
| DecisionStump       | 0.6335   | 0.1951   | 0.7254 |
| sklearn Tree        | 0.9189   | 0.8729   | 0.9258 |

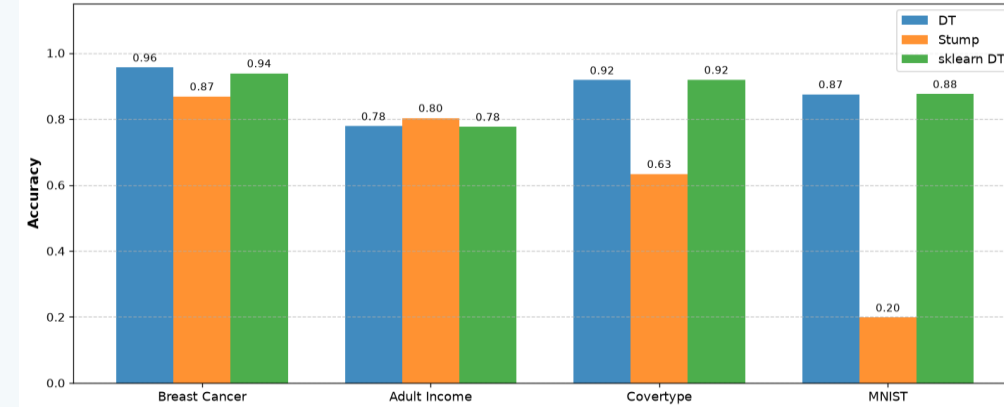
Accuracy difference vs sklearn: 0.13% (PASS)

## 4) MNIST

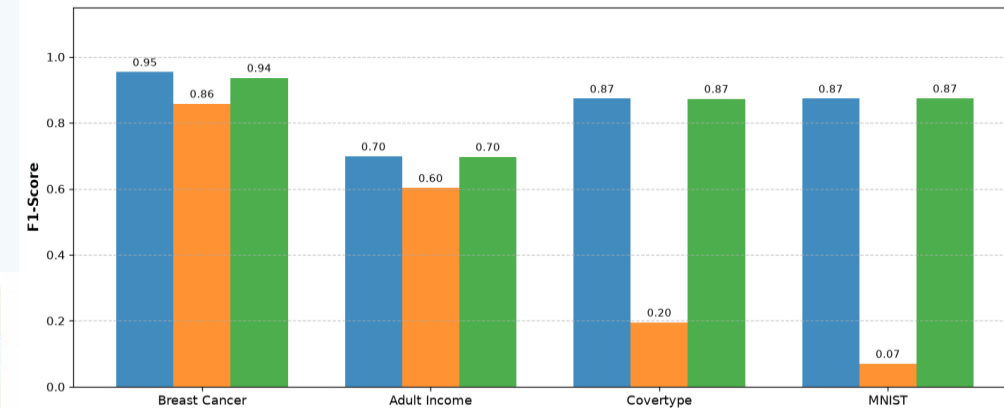
| Model               | Accuracy | F1-score | AUC    |
|---------------------|----------|----------|--------|
| DecisionTree (ours) | 0.8749   | 0.8734   | 0.9298 |
| DecisionStump       | 0.1982   | 0.0697   | 0.6371 |
| sklearn Tree        | 0.8758   | 0.8742   | 0.9302 |

Accuracy difference vs sklearn: 0.10% (PASS)

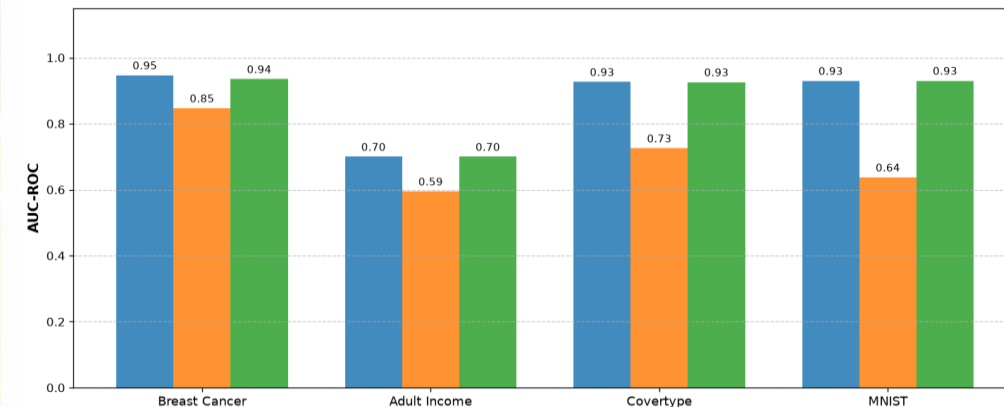
Models Comparison based on Accuracy



Models Comparison based on F1-Score

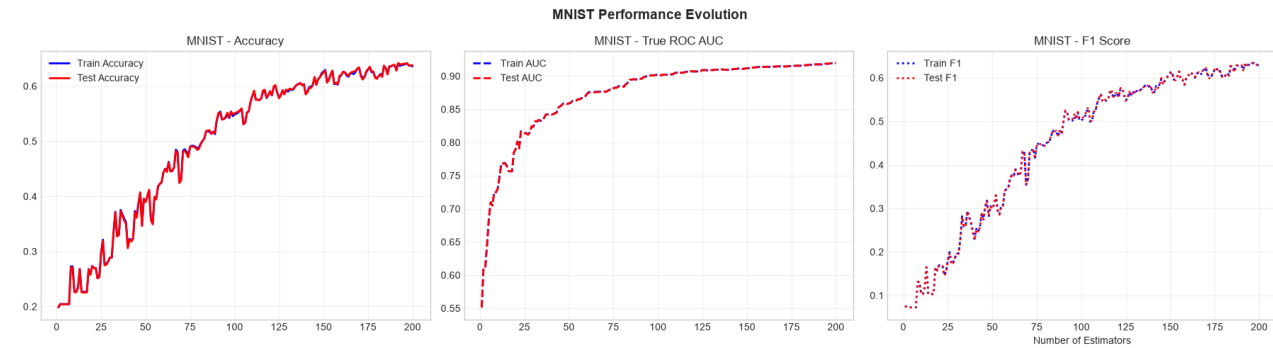
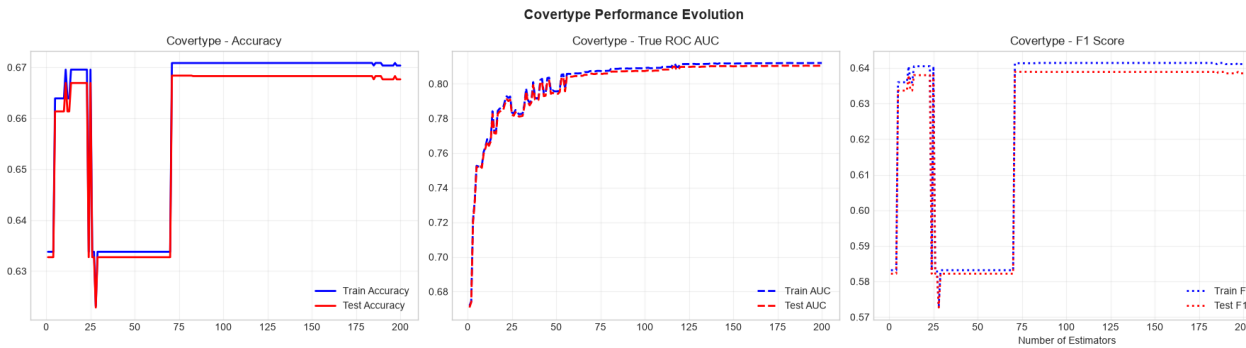
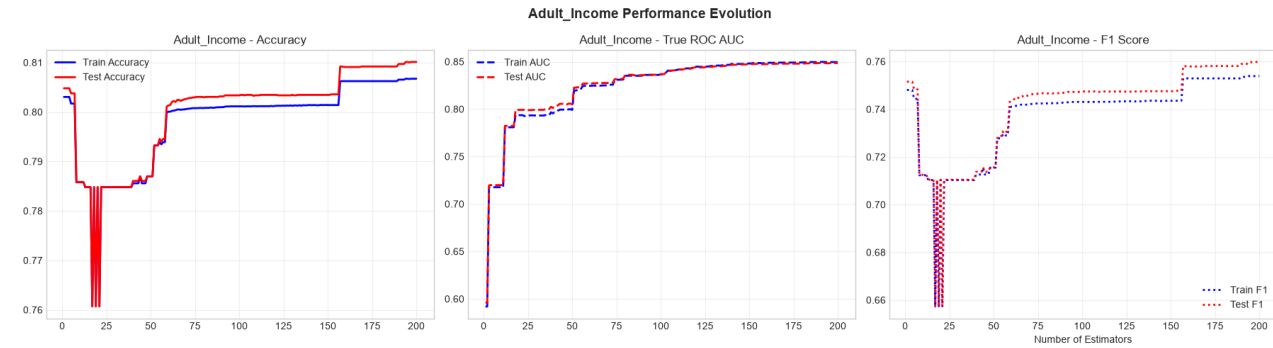
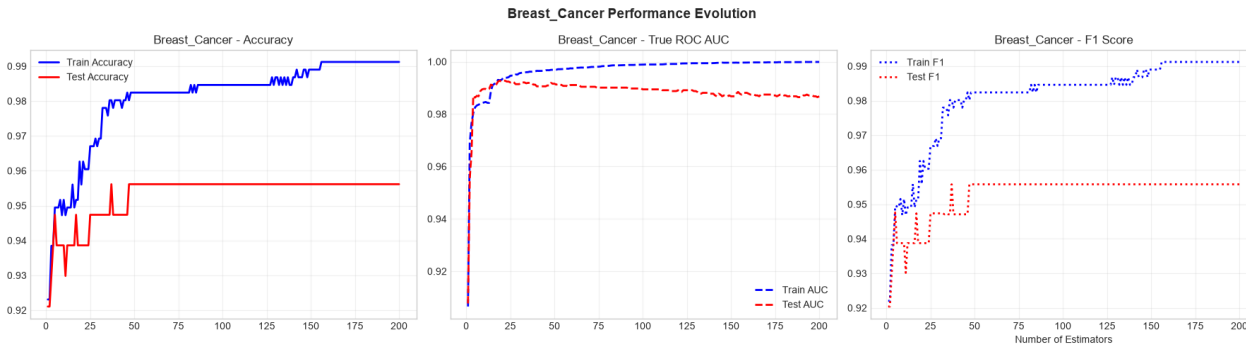


Models Comparison based on AUC-ROC

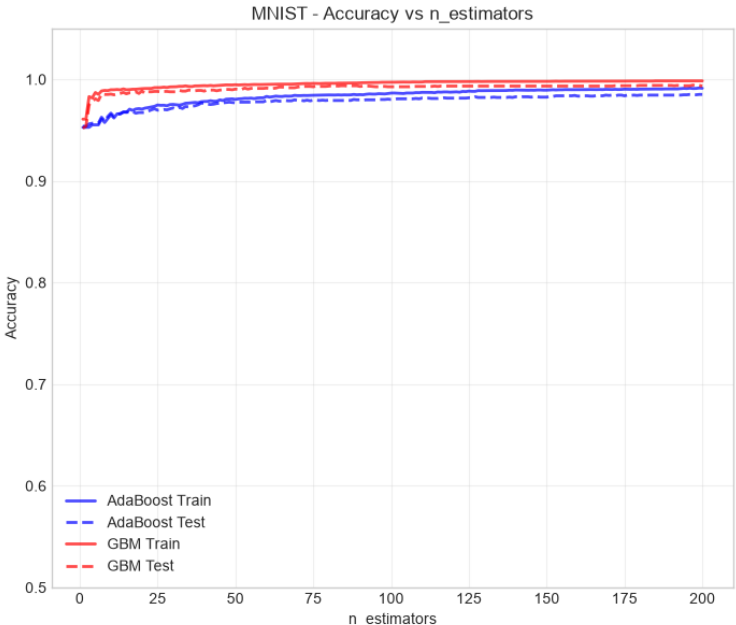
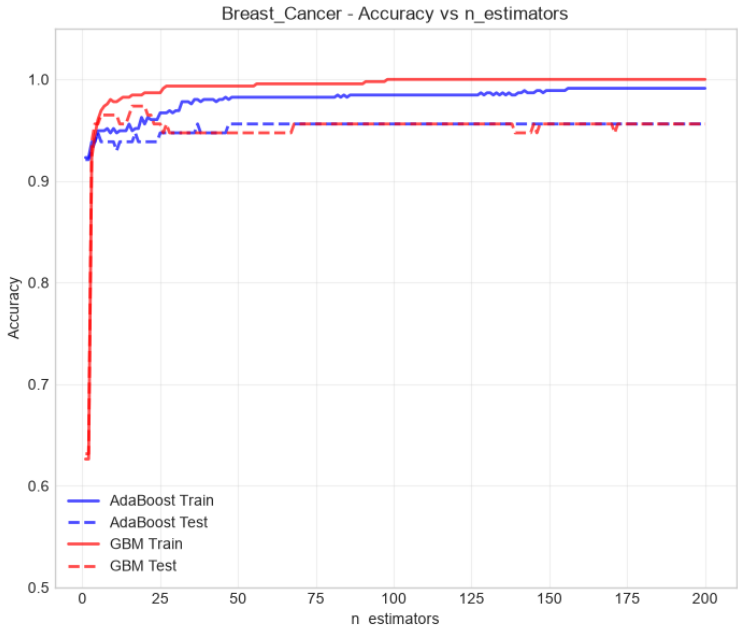
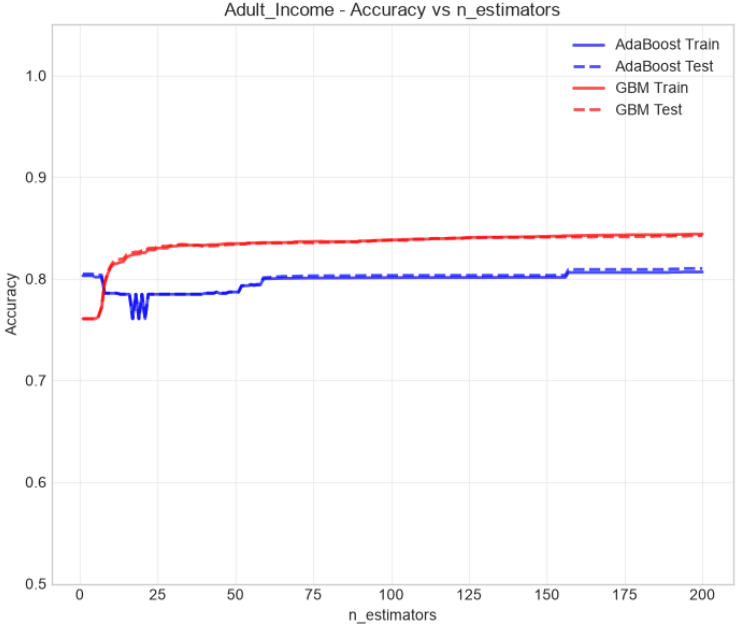
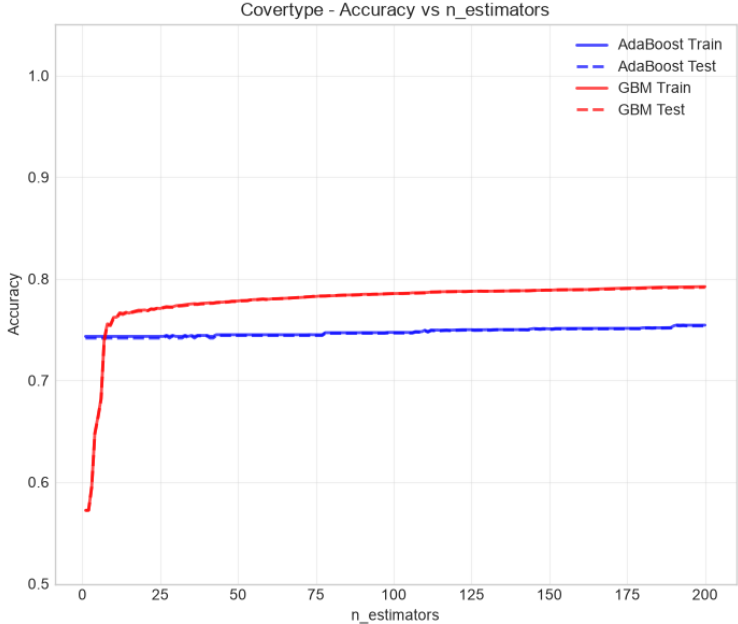




# Adaboost Scaling

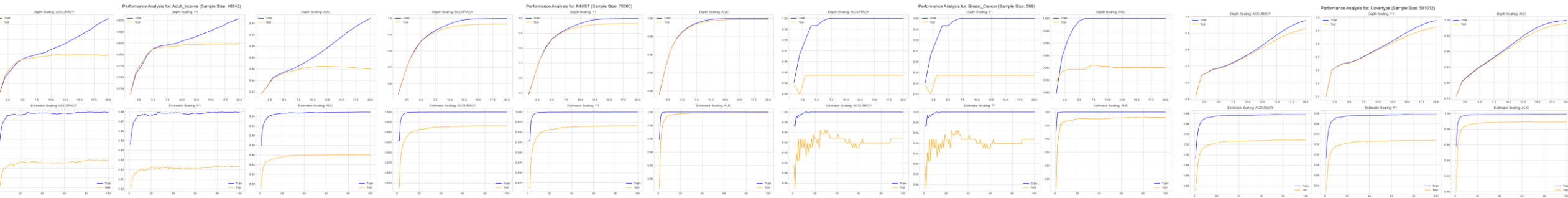
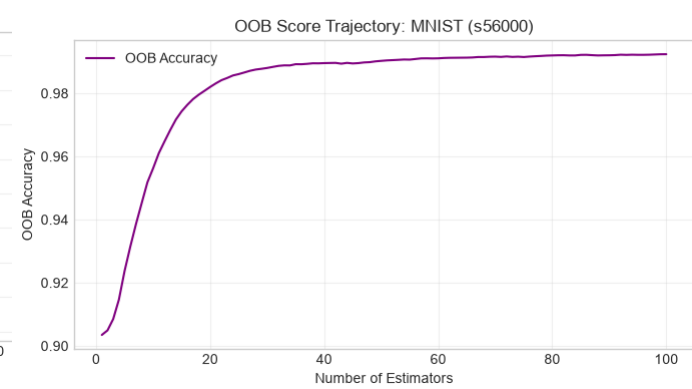
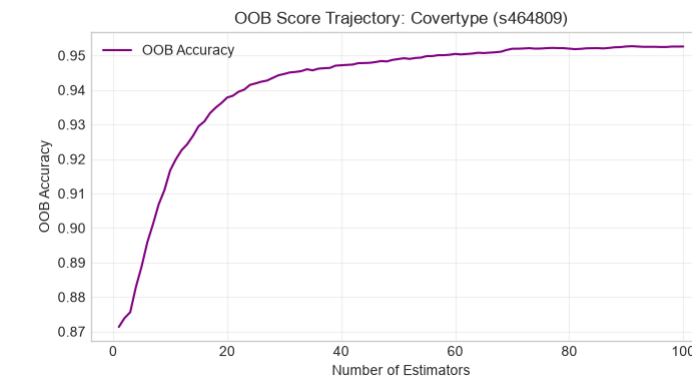
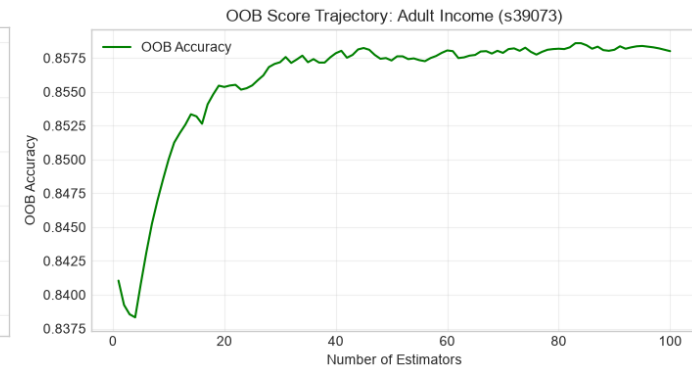
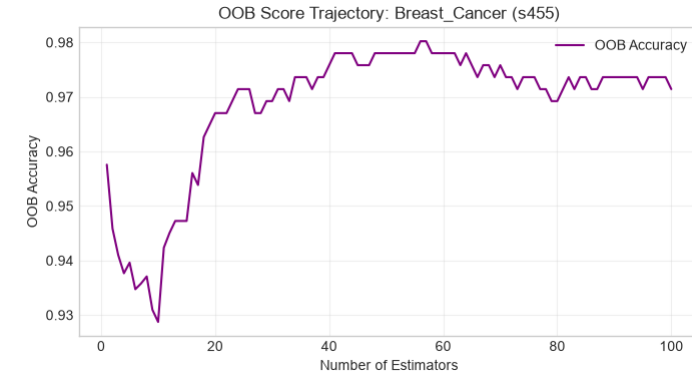
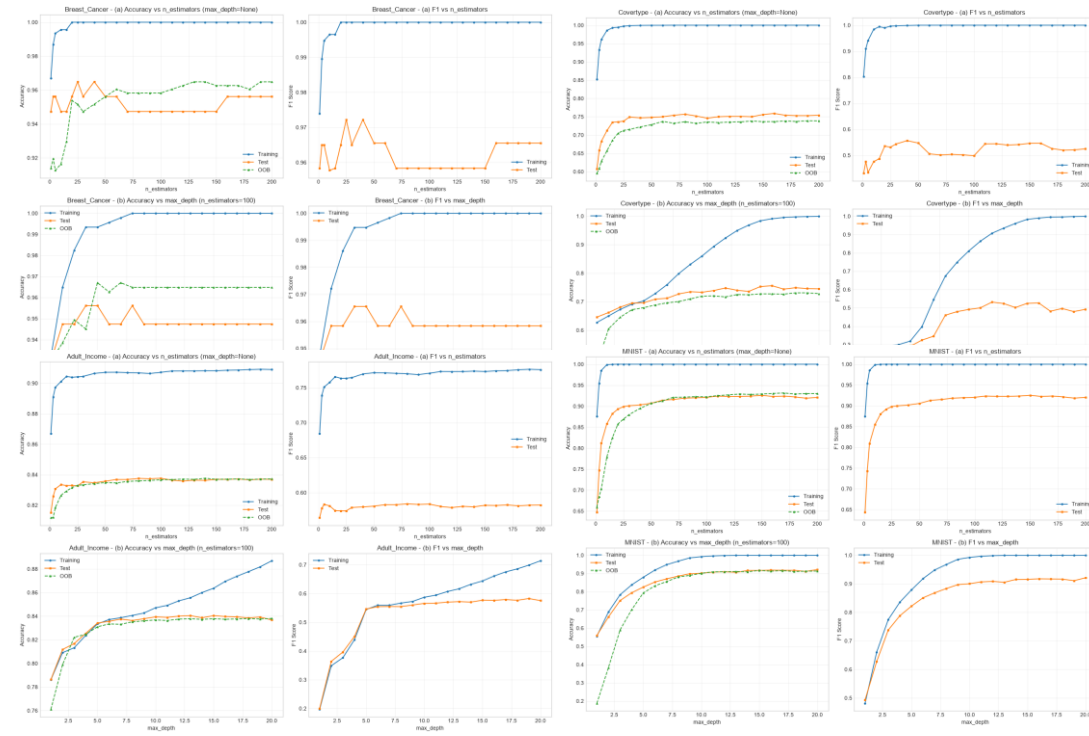


# Gradient Boosting Machine





# Random Forest Scaling





# Head-to-head Comparison



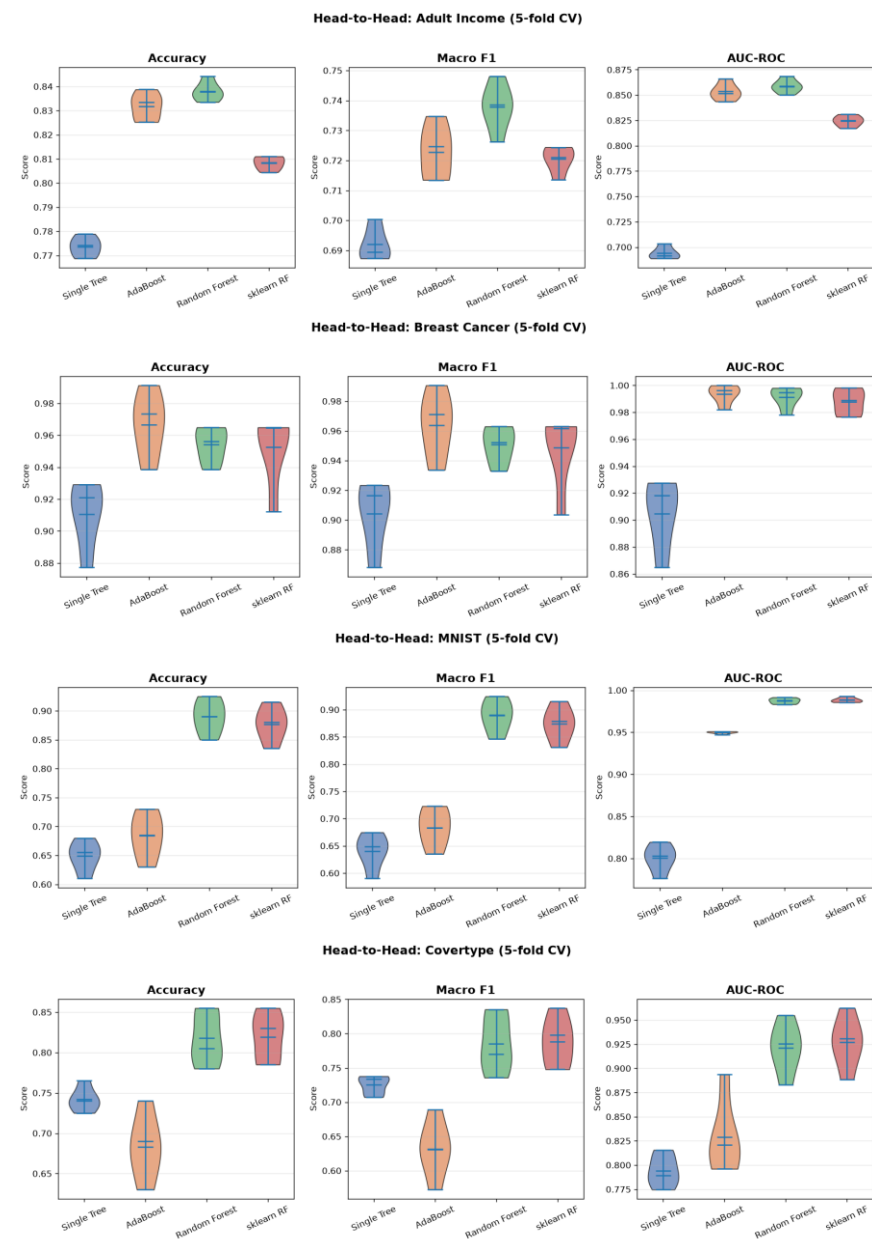
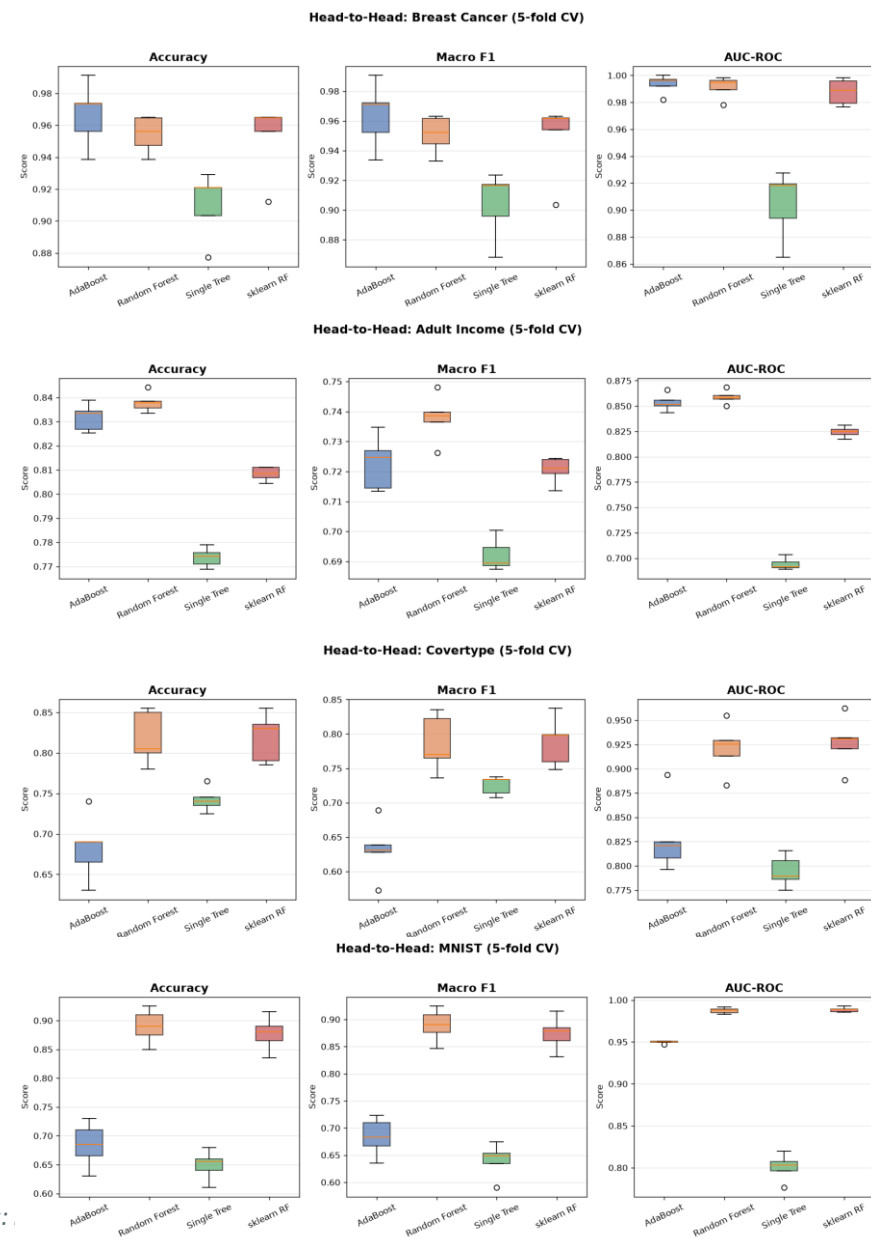
| Adult Income |               |                 |                 |                 |
|--------------|---------------|-----------------|-----------------|-----------------|
| #            | Model         | Accuracy        | Macro F1        | AUC-ROC         |
| 0            | Single Tree   | 0.7737 ± 0.0040 | 0.6920 ± 0.0054 | 0.6942 ± 0.0059 |
| 1            | AdaBoost      | 0.8317 ± 0.0056 | 0.7228 ± 0.0090 | 0.8535 ± 0.0083 |
| 2            | Random Forest | 0.8379 ± 0.0040 | 0.7378 ± 0.0078 | 0.8587 ± 0.0065 |
| 3            | sklearn RF    | 0.8083 ± 0.0028 | 0.7205 ± 0.0044 | 0.8245 ± 0.0053 |

| Breast Cancer |               |                 |                 |                 |
|---------------|---------------|-----------------|-----------------|-----------------|
| #             | Model         | Accuracy        | Macro F1        | AUC-ROC         |
| 0             | Single Tree   | 0.9104 ± 0.0208 | 0.9041 ± 0.0227 | 0.9048 ± 0.0257 |
| 1             | AdaBoost      | 0.9666 ± 0.0200 | 0.9639 ± 0.0217 | 0.9935 ± 0.0070 |
| 2             | Random Forest | 0.9543 ± 0.0114 | 0.9509 ± 0.0125 | 0.9913 ± 0.0080 |
| 3             | sklearn RF    | 0.9526 ± 0.0228 | 0.9489 ± 0.0255 | 0.9878 ± 0.0096 |

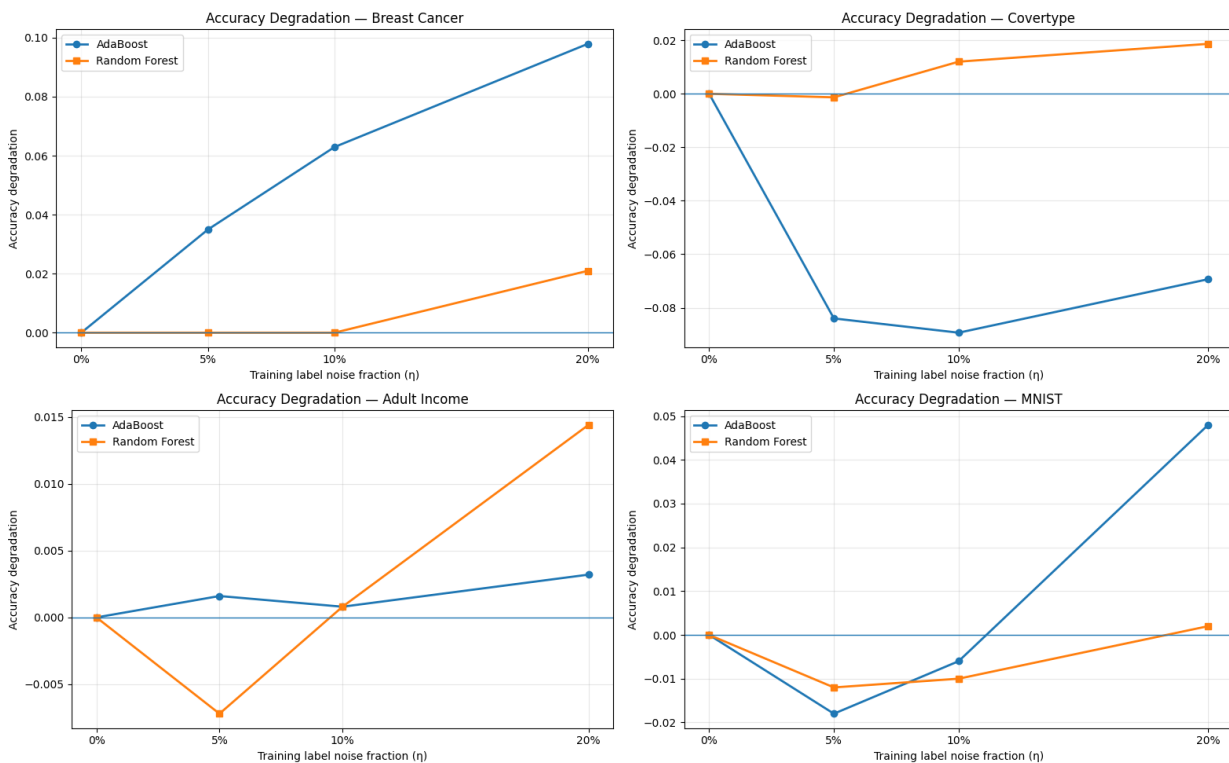
| Covertypes |               |                 |                 |                 |
|------------|---------------|-----------------|-----------------|-----------------|
| #          | Model         | Accuracy        | Macro F1        | AUC-ROC         |
| 0          | Single Tree   | 0.7420 ± 0.0148 | 0.7252 ± 0.0137 | 0.7942 ± 0.0163 |
| 1          | AdaBoost      | 0.6830 ± 0.0402 | 0.6317 ± 0.0414 | 0.8287 ± 0.0381 |
| 2          | Random Forest | 0.8180 ± 0.0329 | 0.7855 ± 0.0418 | 0.9211 ± 0.0261 |
| 3          | sklearn RF    | 0.8190 ± 0.0303 | 0.7884 ± 0.0357 | 0.9269 ± 0.0264 |

| MNIST |               |                 |                 |                 |
|-------|---------------|-----------------|-----------------|-----------------|
| #     | Model         | Accuracy        | Macro F1        | AUC-ROC         |
| 0     | Single Tree   | 0.6490 ± 0.0261 | 0.6402 ± 0.0316 | 0.8005 ± 0.0160 |
| 1     | AdaBoost      | 0.6840 ± 0.0390 | 0.6835 ± 0.0350 | 0.9498 ± 0.0016 |
| 2     | Random Forest | 0.8900 ± 0.0294 | 0.8891 ± 0.0301 | 0.9874 ± 0.0034 |
| 3     | sklearn RF    | 0.8770 ± 0.0297 | 0.8741 ± 0.0312 | 0.9884 ± 0.0028 |

# Head-to-head Comparison



# Noise Robustness



## Model Sensitivity to Label Noise

| # | Dataset                                                                                           | Mean AdaBoost Degradation | Mean Random Forest Degradation | More Sensitive Model                                                                              |
|---|---------------------------------------------------------------------------------------------------|---------------------------|--------------------------------|---------------------------------------------------------------------------------------------------|
| 1 |  Breast Cancer | 0.065268                  | 0.006993                       |  AdaBoost      |
| 2 |  Adult Income  | 0.001867                  | 0.002667                       |  Random Forest |
| 3 |  Covertypes    | -0.080889                 | 0.009778                       |  Random Forest |
| 4 |  MNIST         | 0.008000                  | -0.006667                      |  AdaBoost      |

 Degradation = (Accuracy without noise) – (Average Accuracy with noise at  $\eta$  = 5%, 10%, 20%).  
Higher positive values indicate greater performance drop and thus higher sensitivity to label noise.



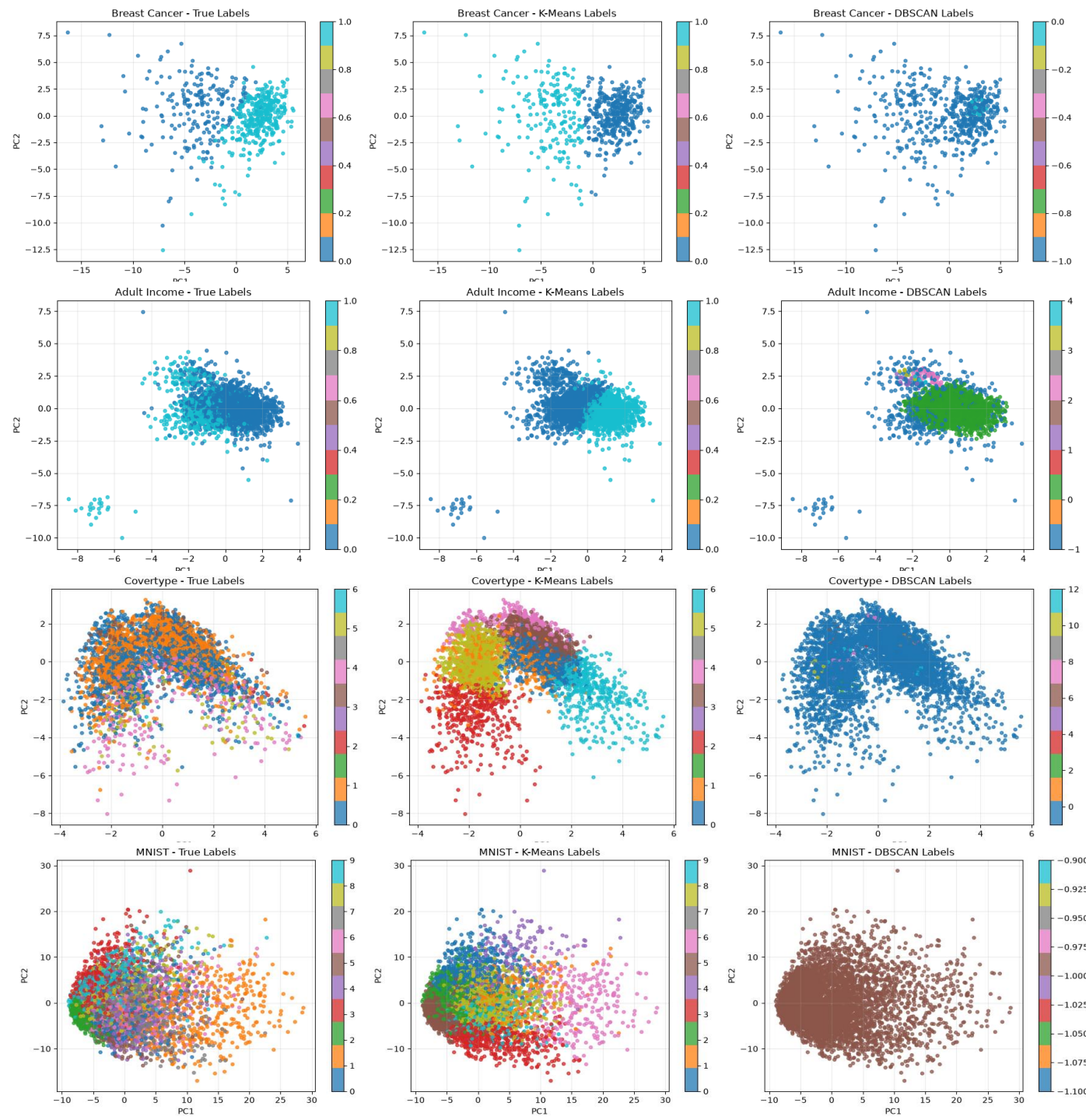
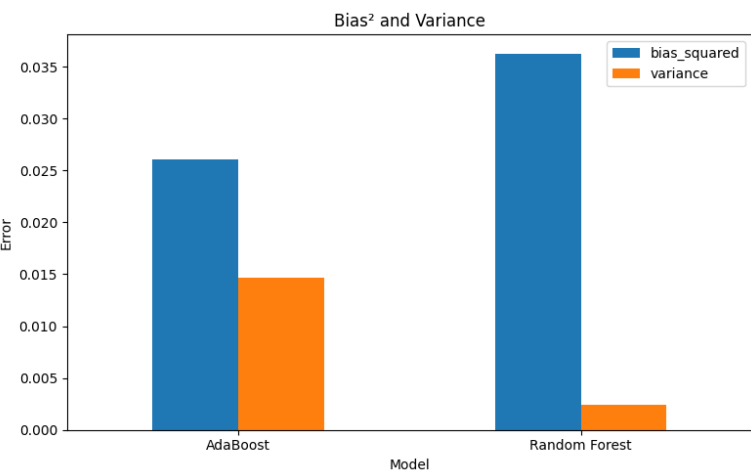


# Unsupervised Analysis

2D PCA scatter plots colored by true class, K-Means clusters, and DBSCAN clusters.

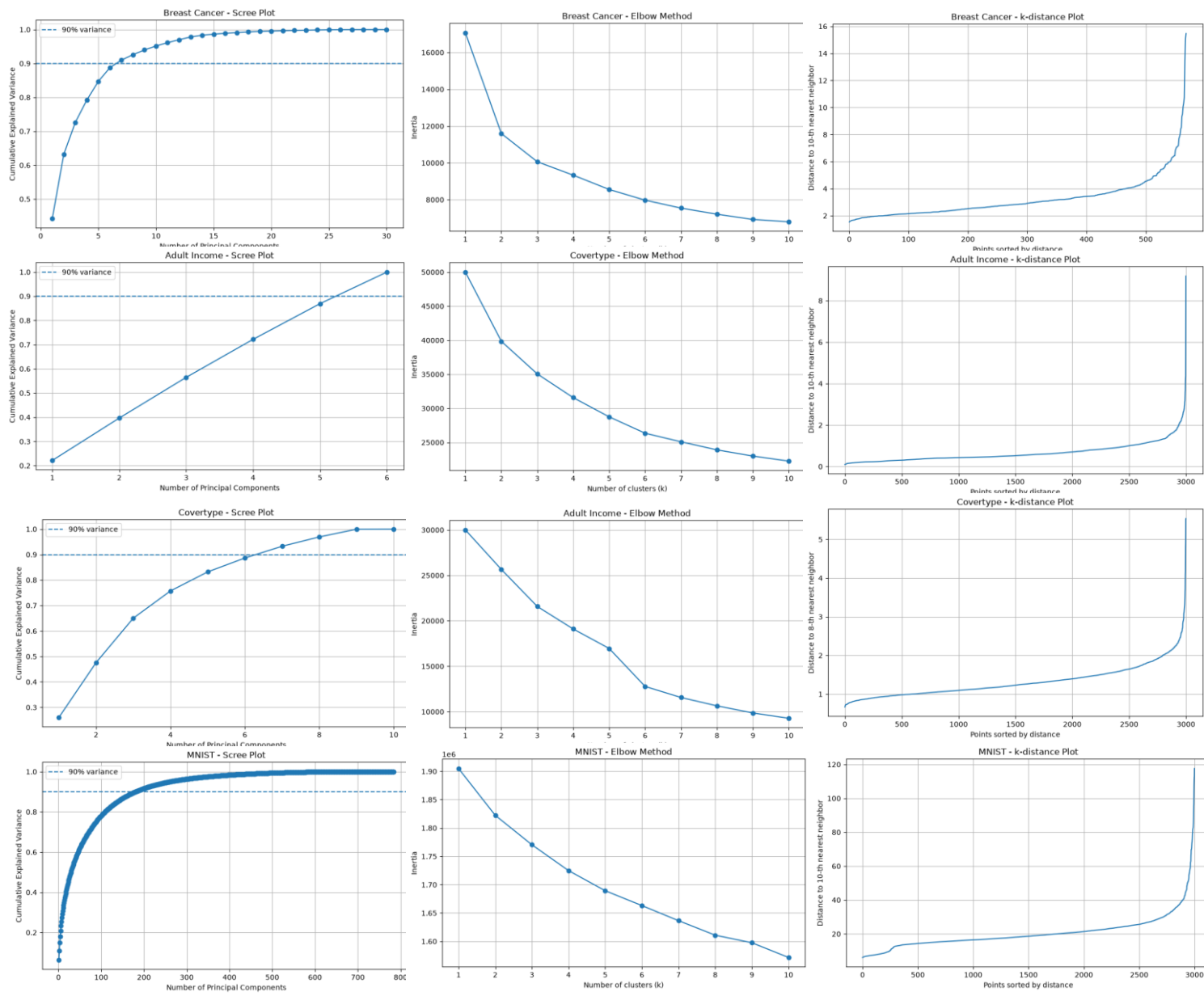
# Bias-variance Decomposition

| Bias-Variance Analysis — Breast Cancer Dataset                                     |          |               |
|------------------------------------------------------------------------------------|----------|---------------|
| Metric                                                                             | AdaBoost | Random Forest |
| Bias <sup>2</sup>                                                                  | 0.026062 | 0.036277      |
| Variance                                                                           | 0.014658 | 0.002404      |
| Total error                                                                        | 0.040720 | 0.038681      |
| Bias <sup>2</sup> + Variance                                                       | 0.040720 | 0.038681      |
| Decomposition difference                                                           | 0        | 0             |
| Mean individual accuracy<br><small>(Average accuracy of individual models)</small> | 0.9556   | 0.9415        |
| Aggregated accuracy<br><small>(Accuracy of the aggregated prediction)</small>      | 0.9649   | 0.9474        |
| All values are computed on the Breast Cancer dataset using 100 resampled datasets. |          |               |





# Unsupervised Analysis



## PCA Variance Explained Summary

| Dataset                                                                                           | PC1 Variance | PC2 Variance | First 2 PCs Total Variance | # of PCs for $\geq 90\%$ Variance |
|---------------------------------------------------------------------------------------------------|--------------|--------------|----------------------------|-----------------------------------|
|  Breast Cancer | 0.4427       | 0.1897       | 0.6324                     | 7                                 |
|  Adult Income  | 0.2217       | 0.1750       | 0.3967                     | 6                                 |
|  Covertypes    | 0.2602       | 0.2155       | 0.4757                     | 7                                 |
|  MNIST         | 0.0632       | 0.0450       | 0.1082                     | 180                               |

**i** PC1 Variance and PC2 Variance represent the proportion of total variance explained by the first and second principal components. First 2 PCs Total Variance is the cumulative variance explained by the first two components. # of PCs for  $\geq 90\%$  Variance is the minimum number of principal components needed to explain at least 90% of the total variance.

## K-Means Clustering Summary ( $k = 1-10$ )

| Dataset                                                                                           | Best ARI $\uparrow$               | Best $k$ (by ARI) |
|---------------------------------------------------------------------------------------------------|-----------------------------------|-------------------|
|  Breast Cancer | 0.6536<br>( $\approx 0.6536246$ ) | 2                 |
|  Adult Income  | 0.0890<br>( $\approx 0.0890122$ ) | 4                 |
|  Covertypes    | 0.0411<br>( $\approx 0.0411467$ ) | 5                 |
|  MNIST         | 0.2813<br>( $\approx 0.2812668$ ) | 10                |

Best ARI indicates the highest Adjusted Rand Index achieved across  $k = 1-10$ , while Best  $k$  denotes the corresponding number of clusters.

## DBSCAN Clustering Summary

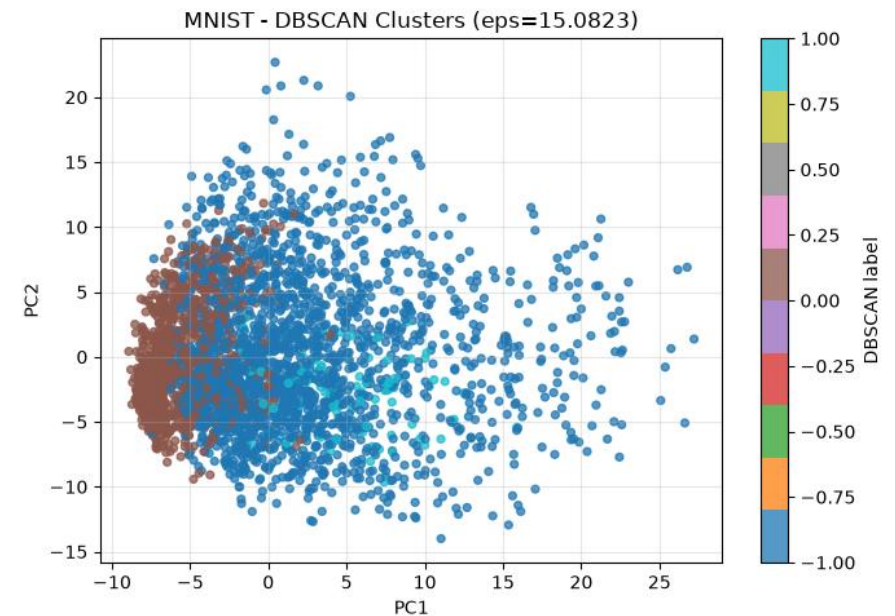
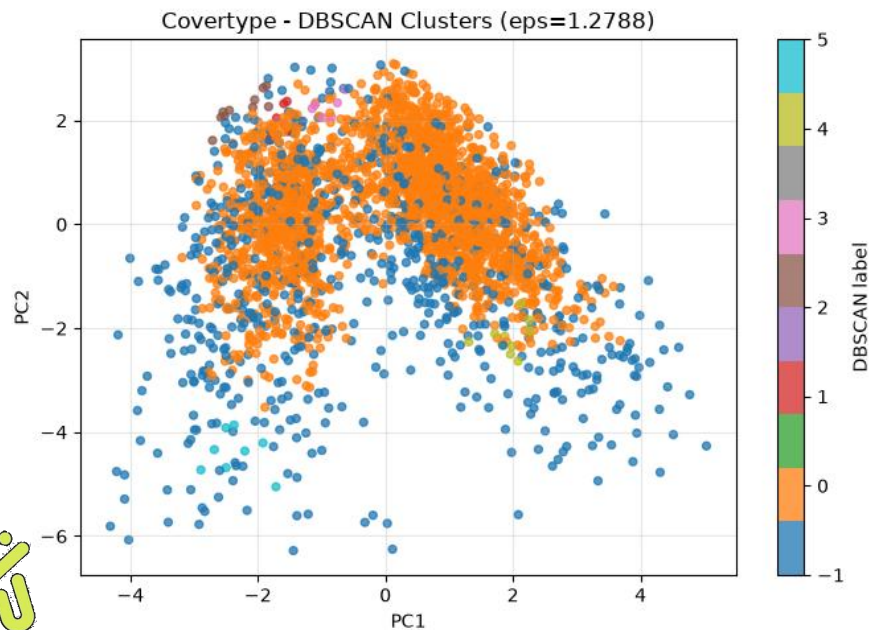
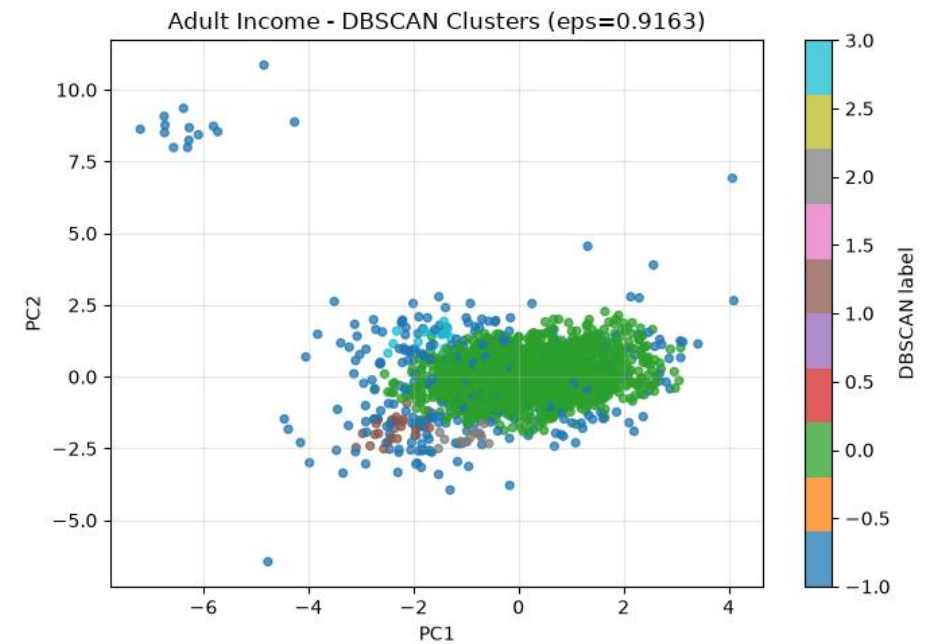
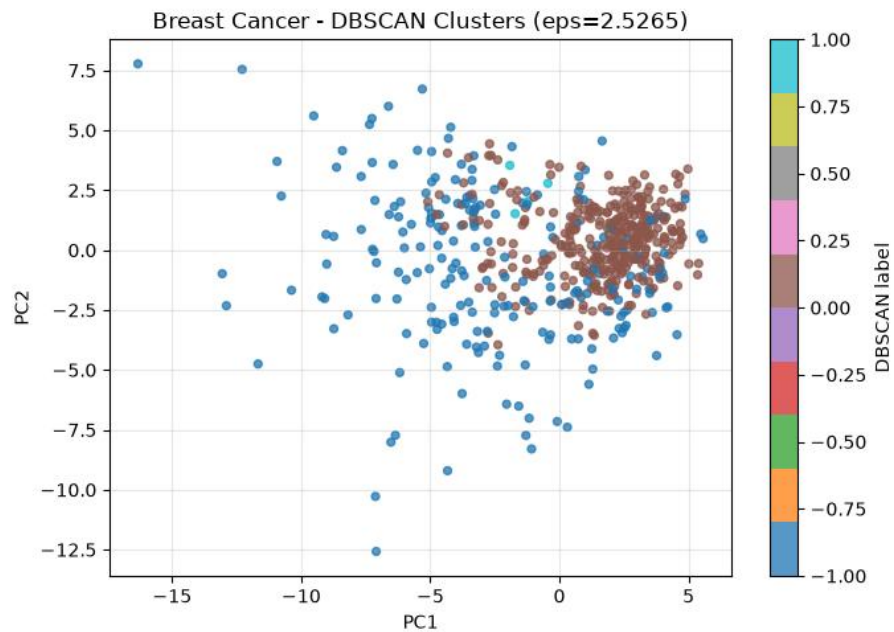
| Dataset                                                                                             | Best $\epsilon$ (Epsilon) | Best ARI (Adjusted Rand Index)      | Clusters (Excluding Noise) | Noise Fraction (Proportion of Noise Points) |
|-----------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------|----------------------------|---------------------------------------------|
|  Breast Cancer | 2.5265                    | 0.271465<br>( $\approx 0.2714646$ ) | 2                          | 0.4587<br>( $\approx 0.458699$ )            |
|  Adult Income  | 0.9163                    | 0.153203<br>( $\approx 0.1532030$ ) | 4                          | 0.1080<br>( $\approx 0.1080$ )              |
|  Covertypes    | 1.2788                    | 0.049041<br>( $\approx 0.0490406$ ) | 6                          | 0.2507<br>( $\approx 0.250667$ )            |
|  MNIST         | 15.0823                   | 0.059434<br>( $\approx 0.0594342$ ) | 2                          | 0.6597<br>( $\approx 0.659667$ )            |

**i** Best  $\epsilon$  is the epsilon value that yields the highest Adjusted Rand Index (ARI). Noise Fraction represents the proportion of points labeled as noise (label = -1) by DBSCAN.



# Unsupervised Analysis

## DBSCAN Plot





# Limitations

- Custom implementations are slower than optimized libraries such as scikit-learn.
- MNIST require high computational time and memory.
- Limited RAM constrained the size of datasets that could be processed in a single run.
- Some experiments used subsets of large datasets to avoid memory overflow and excessive runtime.
- Hyperparameter tuning was limited due to computational cost.
- PCA performs only linear dimensionality reduction and may miss nonlinear structures.

*Thank You — Questions?*

